

PROFESSIONAL SUMMARY

Results-driven Data Analyst & Analytics Engineer with deep expertise in SQL, PostgreSQL, Python (Pandas, NumPy, Scikit-Learn), and Power BI (DAX, Star Schema Modeling). Proven track record in analyzing large-scale enterprise datasets (31K+ transaction line items, 3.2M+ manufacturing telemetry records), uncovering profitability drivers, optimizing discount margins, and developing interactive executive dashboards. Skilled in statistical hypothesis testing, cohort retention decay matrices, ETL data validation, and translating complex data into actionable business strategies.

TECHNICAL SKILLS

Analytics & BI:	Power BI Desktop, DAX Measures, Star Schema Relational Modeling, Tableau Desktop, Advanced Excel (Power Query, VLOOKUP, Pivots)
Databases & SQL:	PostgreSQL, SQLite, Advanced Window Functions (DENSE_RANK, LAG/LEAD), CTEs, Relational Schema DDL, Query Optimization
Python & Data Science:	Python, Pandas, NumPy, Scipy, Matplotlib, Seaborn, Scikit-Learn (Logistic Regression, Feature Engineering), ETL Pipelines
Tools & Methods:	Jupyter Notebook, Git, GitHub, FastAPI (Data APIs), REST APIs, Data Cleaning & Validation, Cohort Analysis, Statistical Modeling

PROFESSIONAL EXPERIENCE

- Composio Technologies Private Limited — Data Validation & Analytics Engineering

Sept 2025 – May 2026

 - Engineered automated data validation pipelines and ETL workflows, verifying schema integrity, JSON payload consistency, and anomaly detection across high-volume event streams.
 - Performed systematic root-cause analysis on data pipeline failures, reducing ingestion discrepancies by 24% and ensuring high data consistency across downstream analytics dashboards.
- MuSigma Inc. — Trainee Decision Scientist / Data Analyst

July 2025 – Aug 2025

 - Leveraged Python (Pandas) and SQL for relational data extraction, cleaning, multi-table joins, aggregation, and Exploratory Data Analysis (EDA) on large-scale business datasets.
 - Built optimized analytical SQL queries and automated reporting transformation workflows, identifying key operational patterns and saving 30+ manual reporting hours weekly.

FEATURED PROJECTS

- E-Commerce Sales & Customer Analytics | Python, PostgreSQL, Power BI, DAX, Seaborn

[GitHub Code](#)

 - Engineered an end-to-end analytics pipeline analyzing 31,855+ transactions (\$8.35M GMV) across 8,500 customers to evaluate product profitability, customer segmentation, discount elasticity, and fulfillment logistics.
 - Designed a normalized PostgreSQL Star Schema with 20+ analytical window queries (DENSE_RANK, LAG/LEAD) and built a Power BI dashboard with 25+ custom DAX measures (YTD, MoM%, Customer LTV Deciles).
 - Empirically identified that discounts >15% eroded gross profit margin from 31.4% to 14.2% without volume surge, and proved repeat buyers generate 59.2% of total profit.
- Manufacturing Quality & Operations Analytics | Python, PostgreSQL, Power BI, DAX, Scipy, IoT

[GitHub Code](#)

 - Analyzed 8,784 production shift records (3.29M manufactured units) to optimize Overall Equipment Effectiveness (OEE), resolve machine downtime root causes, and correlate IoT sensor telemetry with defect generation.
 - Proved strong statistical correlation (r = 0.58) between spindle vibration (>3.0 mm/s) and machining scrap spikes; conducted Pareto downtime analysis showing Tool Wear accounted for 38.4% of plant stoppage hours.
 - Built Power BI plant scorecards tracking First-Pass Yield (96.78%), scrap defect share (3.22%), and machine energy intensity (1.114 kWh/unit).
- Customer Churn & Retention Analytics | Python, Scikit-Learn, PostgreSQL, Power BI, DAX

[GitHub Code](#)

 - Analyzed 25,000 SaaS subscribers (\$23.37M LTV) to uncover churn drivers, visualize cohort retention decay, and protect \$8.14M in Annualized Revenue at Risk (ARR).
 - Created dynamic Monthly Cohort Retention Heatmaps; conducted Chi-Square tests proving Month-to-Month contracts carry 42.1% churn vs 4.6% for 2-Year plans (p < 0.001) and identified support friction cliff (>=4 tickets = 64.7% churn).
 - Developed early-warning customer risk classifier using Logistic Regression (0.84 ROC-AUC) integrated with Power BI retention model.
- Noctra Grid Relay – Full-Stack Spreadsheet Automation & Analytics | Python, FastAPI, Pandas, PostgreSQL, React

[GitHub Code](#)

 - Developed a spreadsheet automation and analytics web application enabling users to upload, preview, clean, validate, and process complex tabular datasets (Excel/CSV) with instant browser rendering.
 - Engineered high-throughput asynchronous FastAPI data endpoints integrated with PostgreSQL and Pandas for automated data cleaning, validation pipelines, and dynamic analytics report generation.

EDUCATION & CERTIFICATIONS

- Bachelor of Engineering (B.E.) | Chandigarh University

Aug 2021 – May 2025 | CGPA: 7.87 / 10.0

Class XII (HBSE): 89.0% | Class X (CBSE): 70.2%
- IBM Data Science Professional Specialization (16-Course Track)
 - Data Visualization with Tableau Specialization (UC Davis, USA)
 - Python for Data Science & AI Certificate (IBM / Coursera)
 - Probability & Statistics for Data Analytics (IIT Kharagpur / NPTEL)